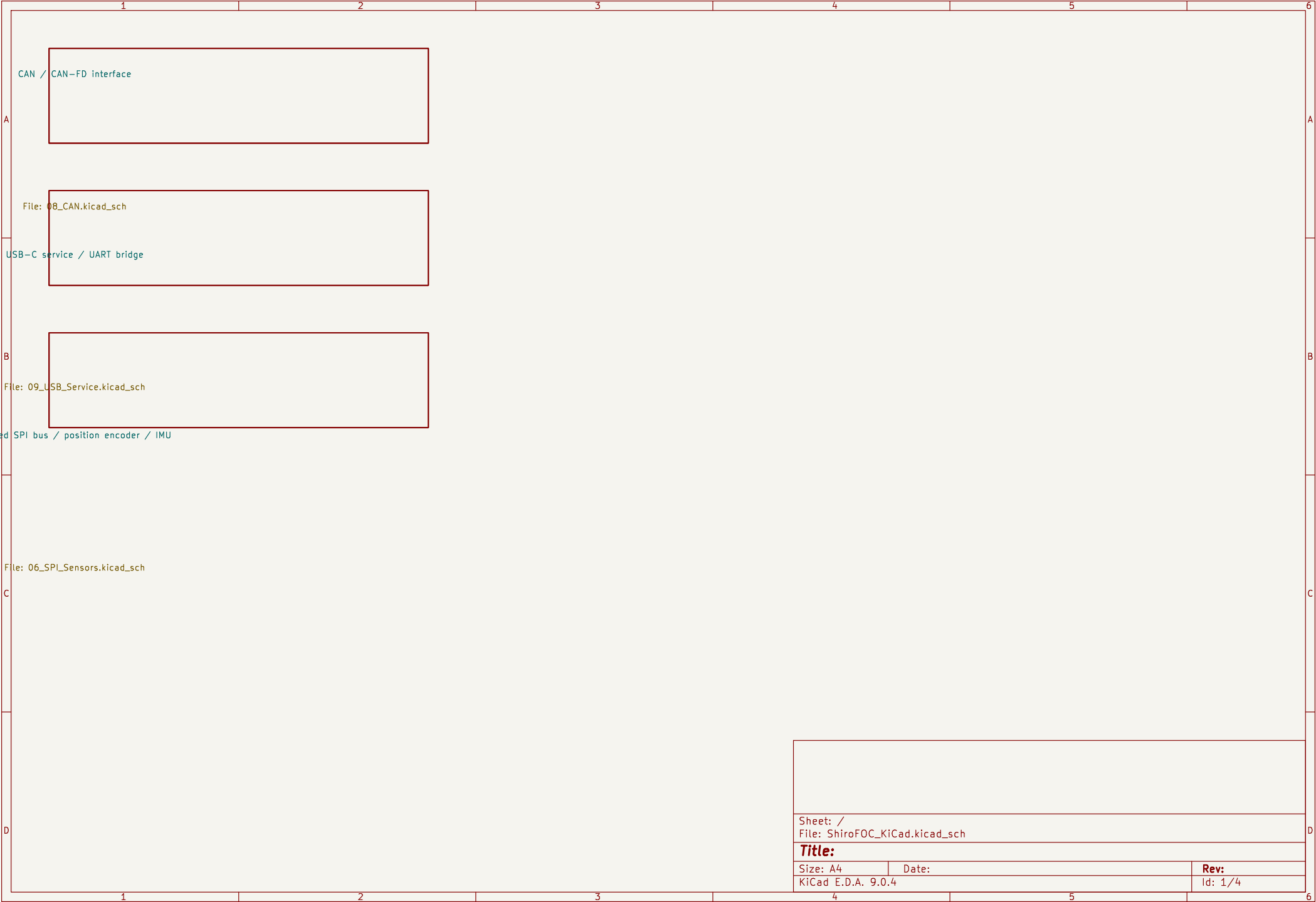
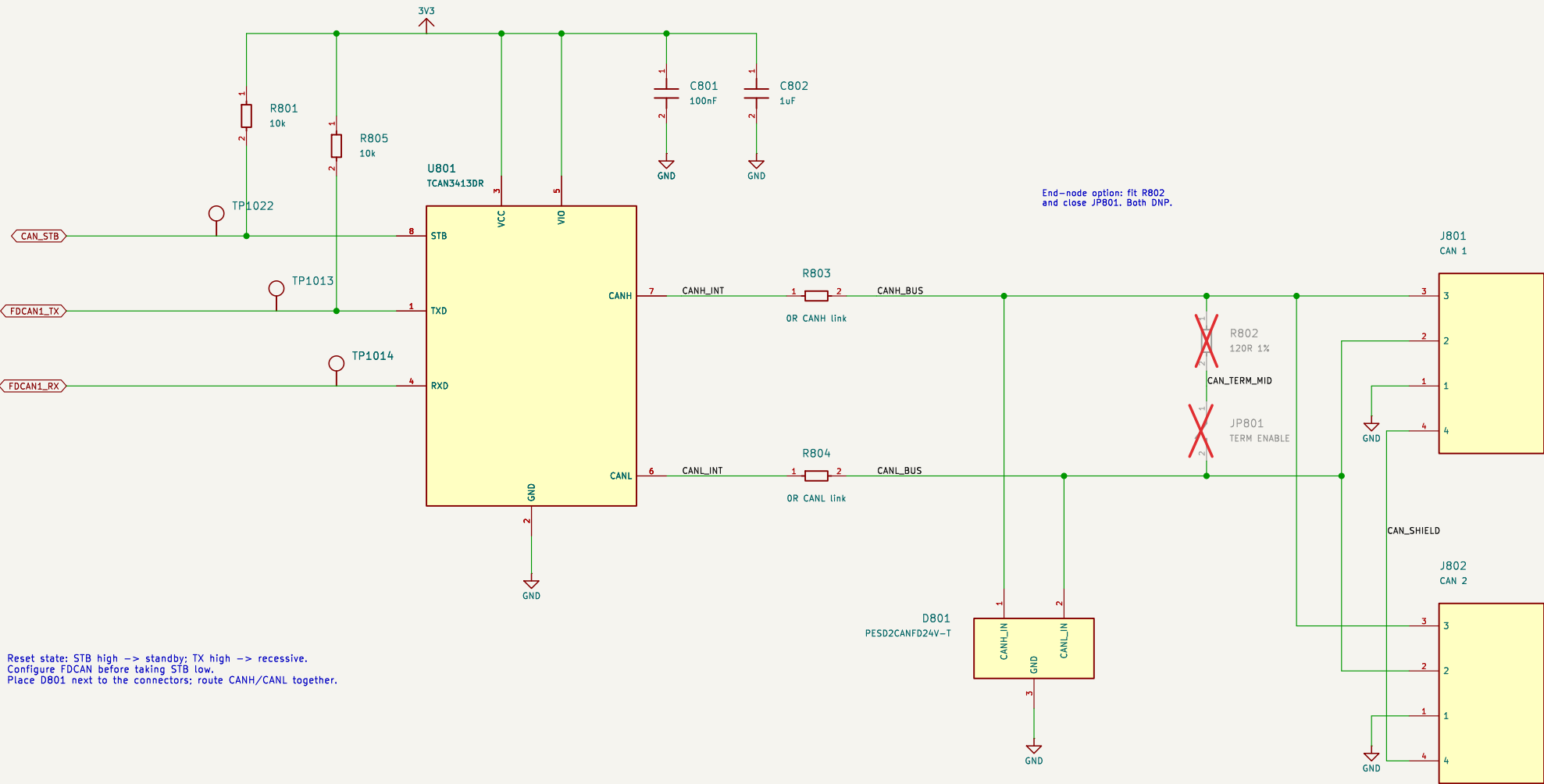




CAN: controller -> transceiver -> protected daisy-chain bus



Reset state: STB high -> standby; TX high -> recessive.
Configure FDCAN before taking STB low.
Place D801 next to the connectors; route CANH/CANL together.

Both headers: 1 GND | 2 CANL | 3 CANH | 4 shield/drain. Pin 4 carries no supply.

Same A0 circuit; connected functional drawing
ShiroFOC

Sheet: /CAN / CAN-FD interface/
File: 08_CAN.kicad_sch

Title: CAN / CAN-FD interface

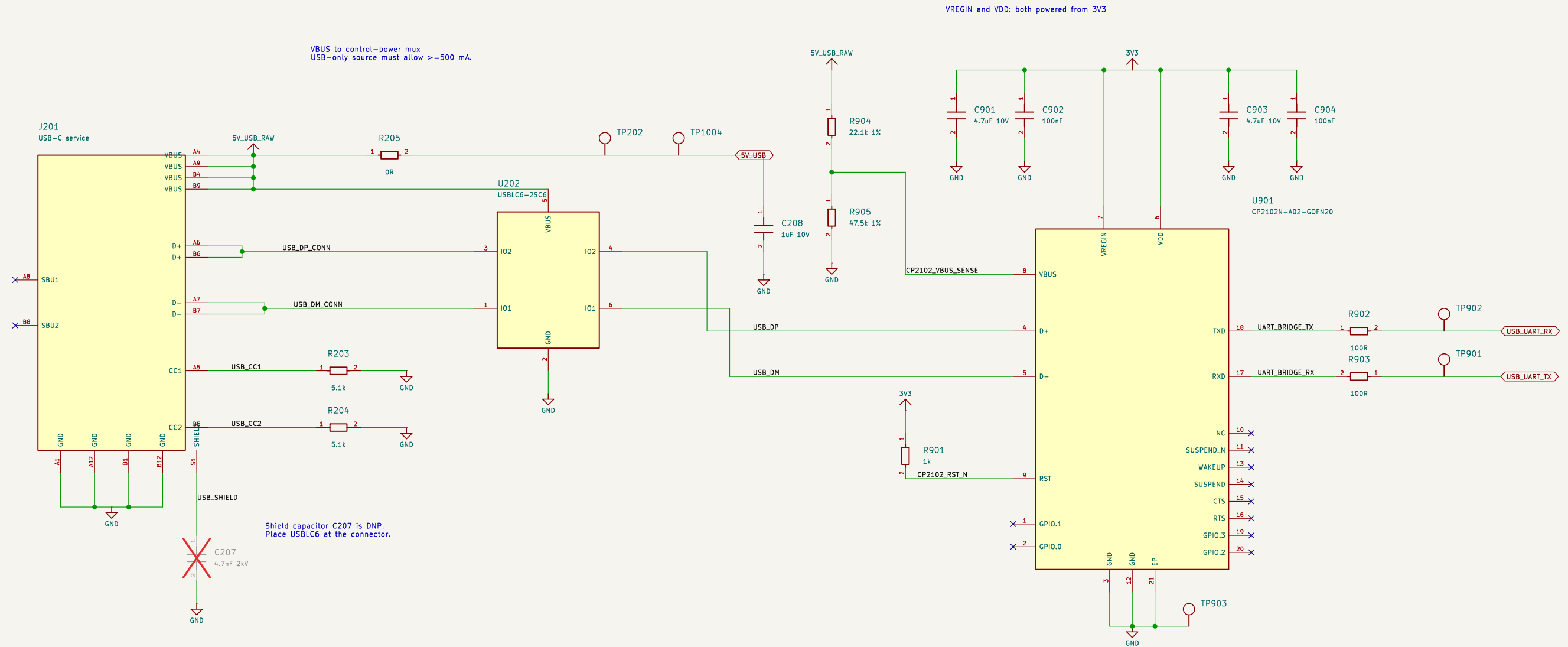
Size: A4 | Date: 2026-09-15

KiCad E.D.A. 9.0.4

Rev: A1-Drawing

Id: 2/4

USB service: connector, protection and bridge in one circuit



LAYOUT: route the USB pair continuously from J201 through U202 to U901; avoid stubs.
C901/C902 belong at VREGIN; C903/C904 belong at VDD. Both pin groups use the same 3V3 rail.

UART labels are named from the MCU point of view.
TXD from U901 drives MCU RX. RTS/CTS are unused.
BOOT0 and reset remain manual controls on the MCU sheet.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /USB-C service / UART bridge/
File: 09_USB_Service.kicad_sch

Title: USB-C service / UART bridge

Size: A3

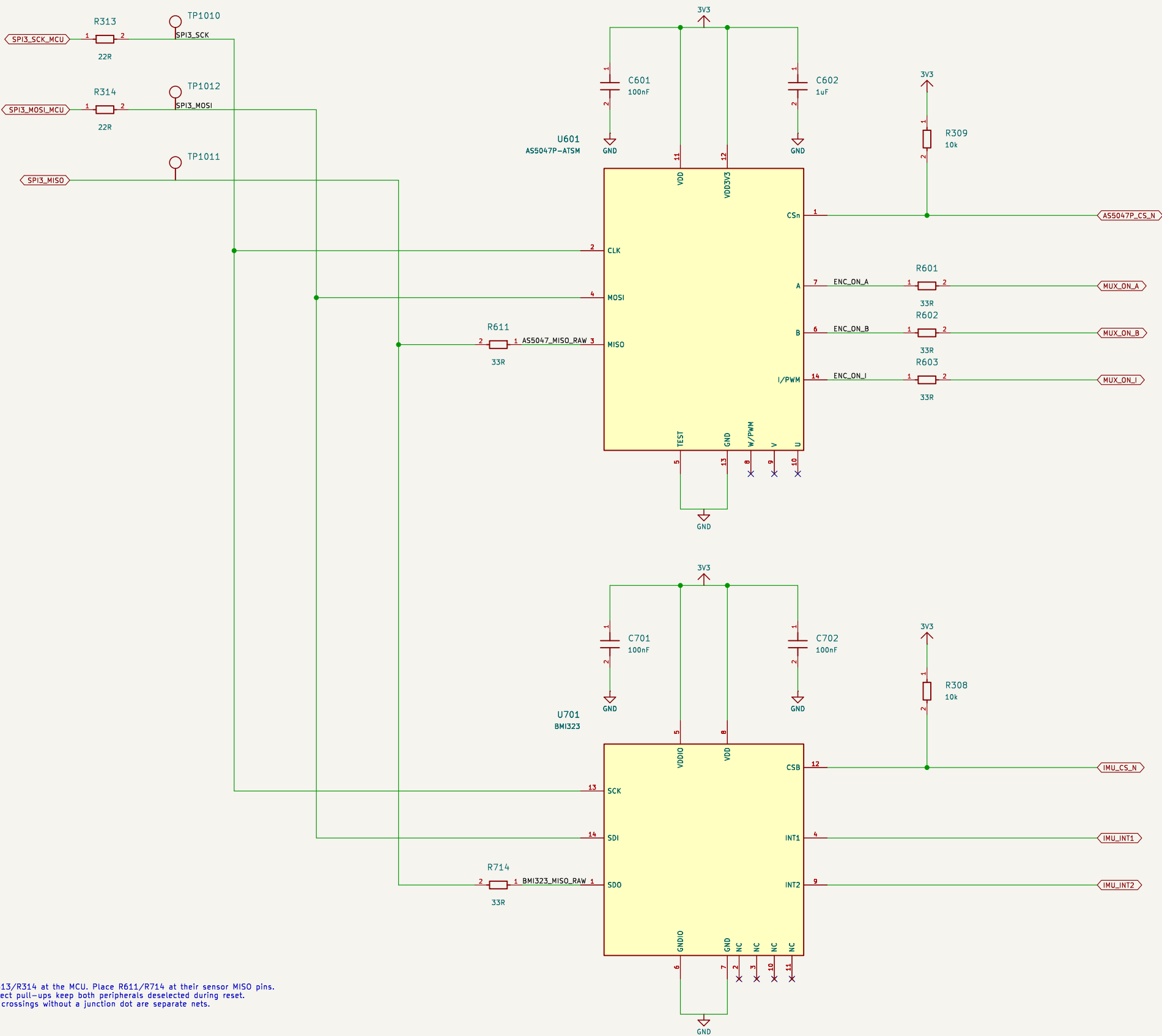
Date: 2026-09-15

Rev: A1-Drawing

Size: A3	
KiCad E.D.A. 9.0.4	

Id: 3/4

SPI3: one shared bus, separate chip selects, local return damping



ONBOARD POSITION
ABI outputs feed the selector sheet.
TEST is tied low for normal operation.
Align the sensor with the shaft magnet.

IMU
Mark X/Y axes on the PCB.
Keep power-stage heat and switching
loops away from the sensor.

Place R313/R314 at the MCU. Place R611/R714 at their sensor MISO pins.
Chip-select pull-ups keep both peripherals deselected during reset.
SPI wire crossings without a junction dot are separate nets.

Same A0 circuit; connected functional drawing		
ShiroFOC		
Sheet: /Shared SPI bus / position encoder / IMU/		
File: 06_SPI_Sensors.kicad_sch		
Title: Shared SPI bus / position encoder / IMU		
Size: A3	Date: 2026-09-15	Rev: A1-Drawing
KiCad E.D.A. 9.0.4		Id: 4/4